Error-Reducing Methodology on the Fast Fourier Transform Option Valuation

March 26, 2012

Abstract

We develop a methodology to reduce the pricing error of the fast Fourier transform (FFT) option valuation approach, a numerical procedure to price vanilla option providing that the characteristic function of the underlying asset's return is known analytically. In our pricing scheme, the option value is decomposed into the proxy and residual terms. The proxy term approximates the theoretical option value and can be analytically evaluated without generating numerical error. The residual term estimates the difference between the theoretical option value and the proxy term and thus generates less numerical error than other methods that price the option directly. Besides, the traditional FFT option valuation approach gives negative pricing results for deep-out-of-the-money options. Our methodology can also solve this problem.

1 Introduction

Various stochastic processes, for example, the jump-diffusion process studied by Merton (1976) and the CGMY model studied by Carr et al. (2002), are proposed to fit the empirical phenomena of the underlying asset's price to value derivative securities more accurately. The martingale valuation approach that prices the options by taking expectation under the risk-neutral measure can be intractable due to complicated density of the underlying asset's return. Fortunately, the characteristic functions of most processes for modelling asset returns are simple and the securities can be numerically evaluated by taking advantages of the Fourier analysis (see e.g. Heston (1993), Bakshi and Madan (2000)). Carr and Madan (1999) introduce the fast Fourier transform (FFT hereafter) into this class of pricing approaches and their work is widely adopted during the past few years.

Carr and Madan (1999) show that the option value can be expressed as the inverse Fourier transform of the Fourier transform of the damped option price, which is a multiplication of the vanilla option price to make sure the Fourier transform is well-defined, and can be numerically determined by the FFT in the inversion stage. However, the integrand of this quadrature is usually highly oscillatory and consequently great computational cost is needed in order to get accurate pricing result. We develop a methodology to further reduce the pricing error and increase the efficiency of the FFT option valuation approach studied by Carr and Madan (1999). Our method assumes that the characteristic function of the underlying asset's return is known analytically, as in the case of Carr and Madan (1999), and that there is another model in which not only the characteristic function of return but the vanilla option value is known analytically. We split the Fourier transform of the damped option price into the proxy and residual parts. The proxy part is chosen as the the Fourier transform of the damped option price implied by a model in which the analytic formula of both the characteristic function of the underlying asset's return and the vanilla option value are admitted, and the residual part is defined as the difference of the two Fourier transforms mentioned above. With this decomposition, the option price is also split into the proxy term, which is the option value implied by the model in which the vanilla option value is admitted, and the residual term, which is defined as the difference of the theoretical option value and the proxy term price. The proxy term can be obtained with an analytic formula and consequently contributes to zero numerical error. Only the residual term has to be evaluated with the FFT and generates numerical error. Therefore, the total pricing error is significantly reduced. In this work, the proxy term of the option value is chosen as the option price implied by the jump-diffusion model studied by Merton (1976) since it can be expressed as a quickly converging series of the option price implied by the diffusion model studied by Black and Scholes (1973). The method of cumulants is used to solve for the corresponding parameters in Merton's jump-diffusion model to get the proxy price as close to the theoretical price as possible.

Furthermore, the original FFT option valuation suggested by Carr and Madan (1999) may generate inaccurate pricing results for deep-out-of-the-money options (see e.g. Carr and Madan (2009)). Our methodology also solves this problem.

The remainder of this paper is organized as follows. Section 2 reviews the option valuation approach studied by Carr and Madan (1999) and the jump-diffusion model studied by Merton (1976). Section 3 introduces how our approach could reduce the pricing error by decomposing the option value into the proxy and residual terms. The way to calibrate the parameters to suppress the differences between the densities for the proxy term and the underlying asset is also discussed in this section. Numerical results in section 4 verify the superiority of our approach. Section 5 concludes.

2 Preliminaries

2.1 The Fast Fourier Transform (FFT)

The FFT is an algorithm which minimizes the number of operations needed for calculating the sum of the following form eq:Sum

$$w(k) = \sum_{j=1}^{N} e^{-i\frac{2\pi}{N}(j-1)(k-1)}\chi(j) \quad \text{for} \quad k = 1, 2, \ldots, N, \tag{1}$$

where N is usually a power of 2 and χ is an analytically-known function. By using the FFT algorithm, the number of operations necessary to obtain the above sum is improved from $O(N^2)$ to $O(N \log_2 N)$.

2.2 The Composite Simpson's Rule

Given an real-valued integrable function f on interval $[a, b]$ with continuous forth derivative and n and even number, there is a $\xi \in [a, b]$ such that eq:Simpson

$$\int_a^b f(x)\,dx = \frac{h}{3}[f(x_0) + 2\sum_{j=1}^{\frac{n}{2}-1} f(x_{2j}) + 4\sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) + f(x_n)] - \frac{b-a}{180}h^4 f^{(4)}(\xi), \tag{2}$$

where $x_j = a + jh$ for $j = 0, 1, ..., n$ with $h = \frac{b-a}{n}$.

The composite Simpson's rule is a method for numerical integration, which approximates definite integrals by

$$\int_a^b f(x)\,dx = \frac{h}{3}[f(x_0) + 2\sum_{j=1}^{\frac{n}{2}-1} f(x_{2j}) + 4\sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) + f(x_n)]. \tag{3}$$

According to Eq. (2), the quadrature error committed by the composite Simpson rule is bounded by eq:error

$$\frac{b-a}{180}h^4 \sup_{y\in[a,b]} \left|f^{(4)}(y)\right|. \tag{4}$$

2.3 Carr and Madan's Option Pricing Method

Let $C_T(k)$ denote the price of a vanilla call option with maturity date T and strike price $K \equiv e^k$. Let $\phi_T(v)$ stand for the characteristic function of the underlying asset's return and $q_T(v)$ denote the risk-neutral density of the logarithm of the underlying asset. Carr and Madan (1999) suggest that eq:IFTpsi

$$C_T(k) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty e^{-ivk}\psi_T(v)\,dv, \tag{5}$$

where $\psi_T(v)$ is the Fourier transform of the damped call price $c_T(k) \equiv e^{\alpha k}C_T(k)$ for some damping constant $\alpha > 0$, which can be rewritten in terms of $\phi_T(v)$ as follows eq:StartPoint, eq:Compute

$$\begin{aligned}
\psi_T(v) &= \int_{-\infty}^{\infty} e^{ivk}c_T(k)\,dk = \int_{-\infty}^{\infty} e^{ivk}e^{\alpha k}C_T(k)\,dk \\
&= \int_{-\infty}^{\infty} e^{ivk}e^{\alpha k}e^{-rT}\int_k^\infty (e^s - e^k)q_T(s)\,ds\,dk \qquad (6)\\
&= e^{-rT}\int_{-\infty}^{\infty} q_T(s)\left[\frac{e^{k(iv+\alpha)+s}}{iv+\alpha} - \frac{e^{k(iv+\alpha+1)}}{iv+\alpha+1}\right]_{k=-\infty}^{k=s} ds \qquad (7)\\
&= \frac{e^{-rT}\,e^{i(v-(\alpha+1)i)\ln S_0}\,\phi_T(v-(\alpha+1)i)}{\alpha^2+\alpha-v^2+i(2\alpha+1)v}.
\end{aligned}$$

The damping constant $\alpha > 0$ is necessary because $C_T(k)$ converges to the spot value of the underlying asset as $k \to -\infty$ and therefore the Fourier transform of $C_T(k)$ does not exist.

Hence we have to define the damped option price such that the Fourier transform of $c_T(k)$ is well defined.

If $\phi_T(v)$ (and hence $\psi_T(v)$) is known analytically, the call value given in Eq. (5) can be approximated by a quadrature eq:CTK

$$C_T(k) \approx \frac{e^{-\alpha k}}{\pi} \sum_{j=1}^{N} e^{-iv_j k} \psi_T(v_j)\eta, \tag{8}$$

where $v_j \equiv \eta(j-1)$; η is sufficiently small and N is large enough to obtain accurate approximation. Based on this setting, the effective upper limit for the integration, which is defined as the truncation point of the integrand, is now equal to $N\eta$.

Since the FFT returns N values of k, we set our values of k as eq:ku

$$k_u \equiv k_1 + \Lambda(u-1) \quad \text{for} \quad u = 1, 2, 3, \ldots, N, \tag{9}$$

where k_1 is an constant such that the interval $[k_1, k_m]$ contains all the contracts in which we are interested.

Substituting Eq. (9) into Eq. (8) yields eq:CTK2

$$\begin{aligned} C_T(k_u) &\approx \frac{e^{-\alpha k_u}}{\pi} \sum_{j=1}^{N} e^{-iv_j(k_1+\Lambda(u-1))} \psi_T(v_j)\eta \\ &= \frac{e^{-\alpha k_u}}{\pi} \sum_{j=1}^{N} e^{-i\Lambda\eta(j-1)(u-1)} e^{-ik_1 v_j} \psi_T(v_j)\eta. \end{aligned} \tag{10}$$

In order to apply FFT, we have to add the constraint eq:LambdaEta

$$\Lambda\eta = \frac{2\pi}{N} \tag{11}$$

to Eq. (10) such that we can invoke the FFT to increase the pricing speed.

The constraint in Eq. (11) implies that if we choose small η in order to increase the pricing accuracy, then we obtain option prices at relatively large strike spacing.

2.4 Merton's Jump-Diffusion Model

In our methodology, the option value under some specific model will serve as a proxy for general pricing. We use the Jump-Diffusion model studied by Merton (1976) to manifest our framework because of the simplicity of its characteristic function as well as first few cumulants. Other suitable processes can be applied without difficulty. Now we give a brief review of Merton's jump-diffusion model.

Assume that an asset price dynamics under some measure $\mathbb{P}$ are eq:SDE

$$dS_t = \left(\mu + \frac{\sigma^2}{2}\right) S_t\, dt + \sigma S_t\, dW_t + (e^{J_t} - 1) S_t\, dN_t, \tag{12}$$

where W_t is a standard Brownian motion; N_t is a Poisson process with parameter λ; J_t is a normally distributed random variable with mean α and standard derivation β.

Applying Itô's formula, we see that the log price process follows eq:lnS

$$d \ln S_t = \mu \, dt + \sigma \, dW_t + J_t \, dN_t \tag{13}$$

and that eq:DiscRate

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}}[S_T] &= S_0 \, \mathbb{E}_{\mathbb{P}}[e^{\mu T}] \, \mathbb{E}_{\mathbb{P}}[e^{\sigma W_T}] \, \mathbb{E}_{\mathbb{P}} \left[\prod_{k=0}^{N_T} J_k \right] \\
&= S_0 \, e^{\mu T} \, e^{\frac{\sigma^2}{2} T} \, e^{\lambda T (e^{\alpha + \frac{\beta^2}{2}} - 1)} \\
&= S_0 \, e^{\left(\mu + \frac{\sigma^2}{2} + \lambda (e^{\alpha + \frac{\beta^2}{2}} - 1) \right) T}.
\end{aligned} \tag{14}$$

According to Eq. (14), the discount rate that must be used under measure $\mathbb{P}$ is

$$\mu + \frac{\sigma^2}{2} + \lambda (e^{\alpha + \frac{\beta^2}{2}} - 1).$$

Furthermore, the return characteristic function implied by the above stochastic differential equation in Eq. (13) is

$$\phi_T^{\text{MJD}}(u) = \exp \left\{ \left(iu\mu - \frac{\sigma^2 u^2}{2} + \lambda \left(e^{i\alpha u - \frac{\beta^2 u^2}{2}} - 1 \right) \right) T \right\},$$

and the first five cumulants of the return are

$$\begin{aligned}
1^{st}\text{cumulant} &= (\mu + \lambda \alpha) T, \\
2^{nd}\text{cumulant} &= \left(\sigma^2 + \lambda \left(\alpha^2 + \beta^2 \right) \right) T, \\
3^{rd}\text{cumulant} &= \lambda (\alpha^3 + 3\alpha\beta^2) T, \\
4^{th}\text{cumulant} &= \lambda (\alpha^4 + 6\alpha^2\beta^2 + 3\beta^4) T, \\
5^{th}\text{cumulant} &= \lambda (\alpha^5 + 10\alpha^3\beta^2 + 15\alpha\beta^4) T.
\end{aligned}$$

The fair price of the vanilla call option with strike price K, maturity T, and underlying asset's price dynamic follows the SDE in Eq. (12) is given by eq:CMJD

$$C_T^{\text{MJD}}(K) \equiv \sum_{j=0}^{\infty} \frac{e^{-\lambda' T} (\lambda' T)^j}{j!} C_T^{\text{BS}}(S_0, K, \sigma_j, r_j), \tag{15}$$

where C_T^{BS} denotes the Black-Scholes call option pricing formula with the initial asset price S_0, the strike price K, the risk-free rate r and the volatility σ.

$$\begin{aligned}
\lambda' &\equiv \lambda e^{\alpha + \frac{\beta^2}{2}}, \\
\sigma_j &\equiv \sqrt{\sigma^2 + \frac{j\beta^2}{T}}, \\
r_j &\equiv \mu + \frac{\sigma^2}{2} + \frac{j(\alpha + \frac{\beta^2}{2})}{T}.
\end{aligned}$$

3 Our Methodology

3.1 Decomposition of the Call Price

We will start from Eq. (5). With a suitable function $\psi_T^{\text{proxy}}(v)$ which closely approximates $\psi_T(v)$, the theoretical call value can be decomposed into a proxy part and a residual part as follows

$$\begin{aligned} C_T(k) &= \frac{e^{-\alpha k}}{\pi}\int_0^\infty e^{-ivk}\psi_T(v)\,dv \\ &= \frac{e^{-\alpha k}}{\pi}\int_0^\infty e^{-ivk}[\psi_T^{\text{proxy}}(v)+\psi_T^{\text{residual}}(v)]\,dv \\ &= \frac{e^{-\alpha k}}{\pi}\int_0^\infty e^{-ivk}\psi_T^{\text{proxy}}(v)\,dv+\frac{e^{-\alpha k}}{\pi}\int_0^\infty e^{-ivk}\psi_T^{\text{residual}}(v)\,dv \\ &= C_T^{\text{proxy}}(k)+C_T^{\text{residual}}(k). \end{aligned} \tag{16}$$

There are two critical criteria for choosing a sound $\psi_T^{\text{proxy}}(v)$. First, the inverse transform of $\psi_T^{\text{proxy}}(v)$ should be analytically known so that $C_T^{\text{proxy}}(k)$ can be obtained without a quadrature. Second, $\psi_T^{\text{proxy}}(v)$ should be analytically known. Without this criterion, it will be very difficult to numerically determine the Fourier inversion of $\psi_T^{\text{residual}}(v)$ because we don't even know the formula of $\psi_T^{\text{residual}}(v)$.

With a $\psi_T^{\text{proxy}}(v)$ satisfies the above two conditions, the proxy term, $C_T^{\text{proxy}}(k)$, can be calculated with an explicit formula, and therefore contributes to zero numerical error. Consequently, all numerical errors come from the residual part, which needs to be obtained from the inverse fast Fourier transform, and the total pricing error will be significantly reduced.

3.2 Use Merton's jump-diffusion model as our proxy

Suppose that there is a target distribution with analytic characteristic function of the return. Since the characteristic function (for a vast class) admits infinite differentiability (see e.g. Bakshi and Madan (2000)), we assume the first five cumulants of the return implied by the target distribution is known and are equal to $m_i,\ i=1,2,3,4,$ and 5. In this work, we choose the Fourier transform of the damped call price implied by the SDE in Eq. (12) as $\psi_T^{\text{proxy}}(v)$, which can be written as

$$\psi_T^{\text{proxy}}(v)=\frac{e^{-(\mu+\frac{\sigma^2}{2}+\lambda(e^{\alpha+\frac{\beta^2}{2}}-1))T}\,e^{i(v-(\alpha+1)i)\ln S_0}\,\phi_T^{\text{MJD}}(v-(\alpha+1)i)}{\alpha^2+\alpha-v^2+i(2\alpha+1)v}.$$

Therefore, the $C_T^{\text{proxy}}(k)$, in Eq. (16) is analytically known and can be written as Eq. (15). The method of cumulants is used for calibrating the parameters to suppress the differences between the two characteristic functions of the return for the analytical term and the underlying asset.

We begin matching the first five return cumulants of Merton's jump-diffusion model with the first five return cumulants implied by the underlying asset's price process, this yields eq:c1, eq:c2, eq:c3, eq:c4, eq:c5

$$
\begin{aligned}
&(\mu + \lambda\alpha)\, T = m_1, && (17)\\
&\left(\sigma^2 + \lambda\left(\alpha^2 + \beta^2\right)\right) T = m_2, && (18)\\
&\lambda(\alpha^3 + 3\alpha\beta^2)T = m_3, && (19)\\
&\lambda(\alpha^4 + 6\alpha^2\beta^2 + 3\beta^4)T = m_4, && (20)\\
&\lambda(\alpha^5 + 10\alpha^3\beta^2 + 15\alpha\beta^4)T = m_5. && (21)
\end{aligned}
$$

Since the time to maturity is a known constant, we set $T = 1$ W.L.O.G. for notation simplicity in the following derivation.

We first rewrite Eq. (19) as eq:beta

$$\beta^2 = \frac{m_3 - \lambda\alpha^3}{3\lambda\alpha} \tag{22}$$

and then replace all the β^2 in Eq. (20) and Eq. (21) by Eq. (22). This yields

$$
\begin{aligned}
&m_3^2 + 4m_3\alpha^3\lambda - \alpha^2\lambda(3m_4 + 2\alpha^4\lambda) = 0,\\
&5m_3^2 - \alpha\lambda(3m_5 + 2\alpha^5\lambda) = 0.
\end{aligned}
$$

By equating the above two equations, we get eq:lambda

$$\lambda = \frac{4m_3^2}{3m_5\alpha - 3m_4\alpha^2 + 4m_3\alpha^3}. \tag{23}$$

Now we can replace the λ and β^2 in Eq. (19) by Eq. (22) and Eq. (23) respectively and get a polynomial equation as follows eq:alpha

$$5m_3^2(3m_5 + \alpha(4m_3\alpha - 3m_4))^2 - 4m_3^2(9m_5^2 + 8m_3^2\alpha^4 + 3m_5\alpha(4m_3\alpha - 3m_4)) = 0. \tag{24}$$

Clearly, the variable α in Eq. (24) can be easily solved since the left hand side of Eq. (24) is a polynomial function of α with degree 4. However, it is possible that Eq. (24) does not have any roots among the real numbers. In this case, we simply minimizes the absolute value of the left hand side of Eq. (24). Once α is determined, the other 4 parameters can also be determined easily.

Figure 1 provides an inspirational example for our method by the Variance Gamma model studied by Madan et al. (1998). The solid line represents the probability density implied by the Variance Gamma model with volatility of the Brownian motion $\sigma = 0.1213$, variance rate of the gamma time change $\nu = 0.1686$, and drift in the Brownian motion with drift $\theta = -0.1436$. The dashed line represents the probability density of Merton's jump-diffusion model, with parameters solved by the method provided in this section.

Figure 2 demonstrate the error-reducing effect of our algorithm by the Variance Gamma model with same parameters as in Figure 1. Since we use the composite Simpson's rule, as suggested in Carr and Madan (1999), for quadrature, the numerical error committed is bounded by a constant multiple of the maximum absolute value of the fourth derivative of the integrand, as in Eq. (4). Hence we demonstrate the integrand of the Fourier as well as

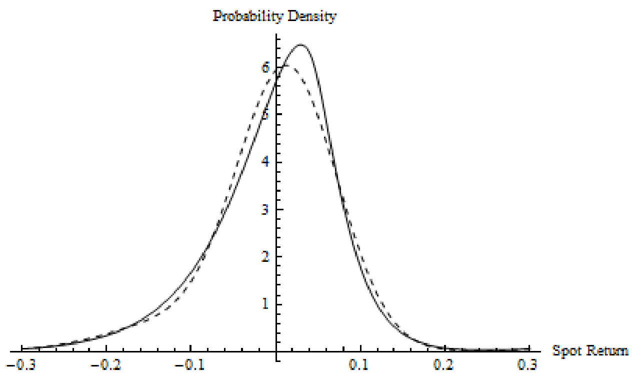


Figure 1: Comparison of the Target and Proxy Densities

The solid line represents the probability density implied by the Variance Gamma model with parameters $\sigma = 0.1213$, $\nu = 0.1686$, and $\theta = -0.1436$. The dashed line represents the probability density of Merton's jump-diffusion model, with parameters solved by the method provided in this section..

its forth derivative. The real part of the integrands of the Fourier inversion in the original Carr and Madan's method and our method are shown in panel (a) and (b), respectively. A comparison of panel (a) and panel (b) is shown is panel (c). The real part of the fourth derivative of the integrands of the Fourier inversion in the original Carr and Madan's method and our method are shown in panel (d) and (e), respectively. A comparison of panel (d) and panel (e) is shown is panel (f). Since the integrand and its fourth derivative are much more smoother and have smaller amplitude in our method than in the original Carr and Madan's method, the total pricing error is significantly reduced.

4 Numerical Results

We take the Variance Gamma model and the Heston model for numerical examples. For the Variance Gamma model, we use the same parameters as in Figure 1. The Heston parameters are mean reversion $\kappa = 1.49$, long-run variance $\theta = 0.0671$, volatility of volatility $\sigma = 0.742$, correlation $\rho = -0.571$, and current variance $V_0 = 0.0262$. The time-error plot is used for comparing the pricing results for the original Carr and Madan's algorithm and our algorithm. The convergence rate for both models are shown in Figure 3. The Given the same computational time constraint, the pricing accuracy for our method are better than the original Carr and Madan's method in both cases. It is because of this that the computational time depends mostly on the number of grid used for quadrature and the number of grids needed for our method is much less than the original Carr and Madan's method due to the smaller amplitude of the integrand in the Fourier inversion. Thus we conclude that our method does significantly improve the pricing efficiency.

Furthermore, the original Carr and Madan's method often returns negative prices for deep out-of-the-money options (see e.g. Carr and Madan (2009)) because the integrand of the Fourier inversion for deep out-of-the-money options is highly oscillatory, and therefore deteriorates the convergence rate of quadrature. Our method can also solve this problem.

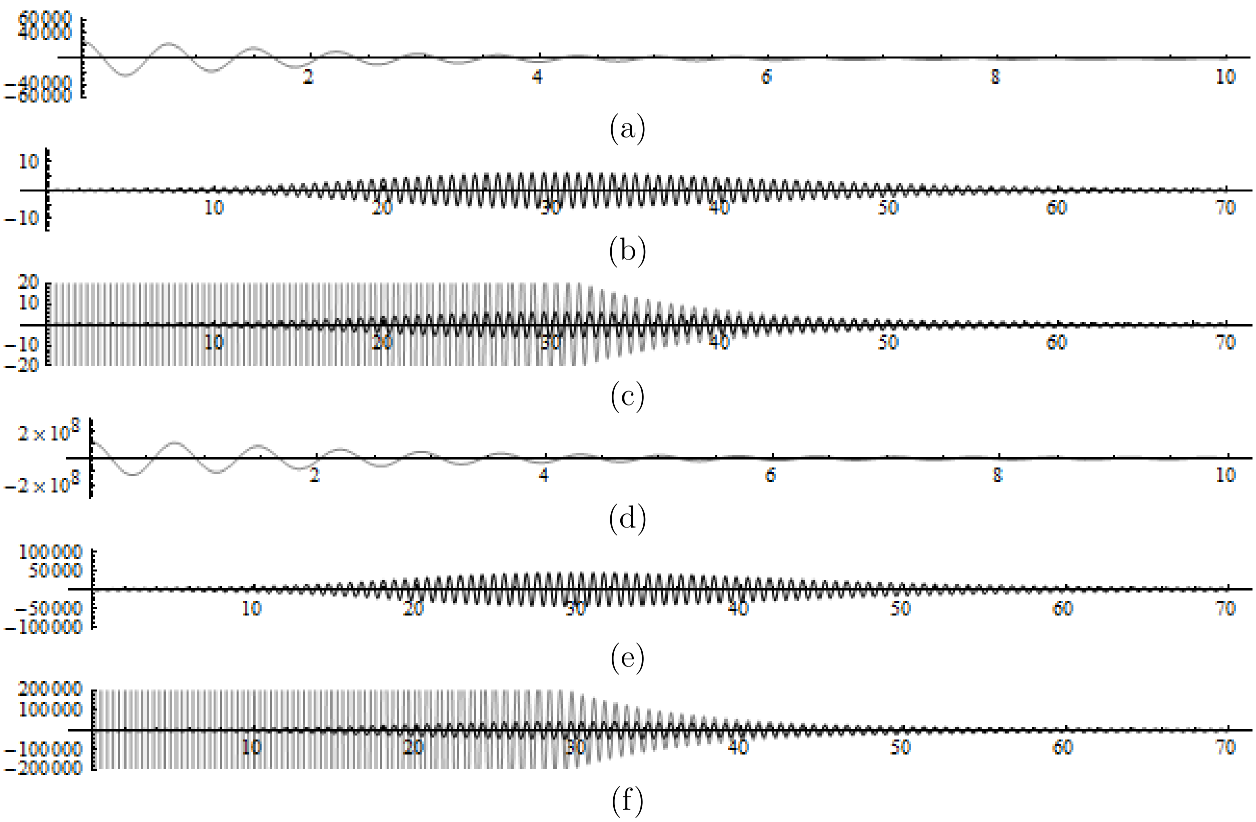
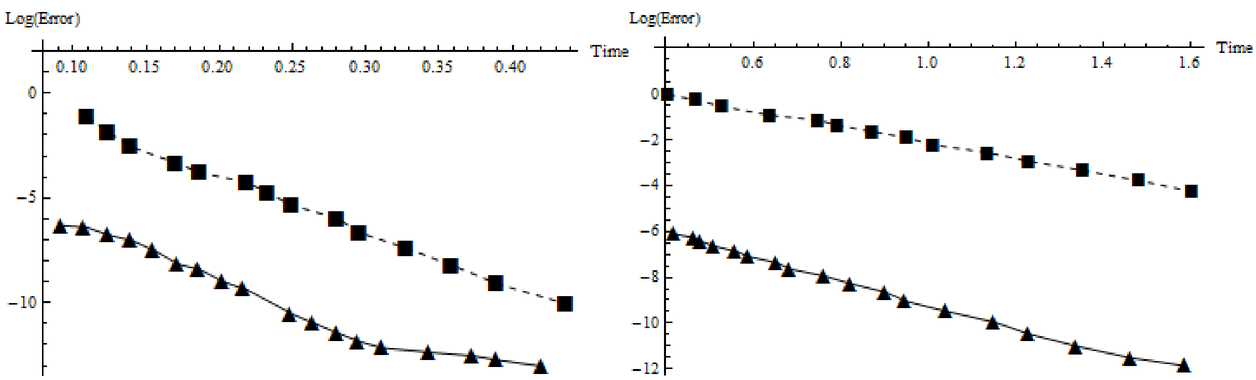


Figure 2: Comparison of the Integrand in the Fourier Inversion

The real part of the integrands of the Fourier inversion in the original Carr and Madan's method and our method are shown in panel (a) and (b), respectively. A comparison of panel (a) and panel (b) is shown is panel (c). The real part of the fourth derivative of the integrands of the Fourier inversion in the original Carr and Madan's method and our method are shown in panel (d) and (e), respectively. A comparison of panel (d) and panel (e) is shown is panel (f). All panels are based on the Variance Gamma model with setting $S_0 = 100$, $K = 100$, $r = 0$, $T =$ 4 months, $\sigma = 0.1213$, $\nu = 0.1686$, and $\theta = -0.1436$.



(a) (b)

Figure 3: Convergence of Pricing Results

The convergence rate of pricing results for the VG model and the Heston model are shown in panel (a) and panel (b), respectively. The x-axis denotes the time used for pricing and the y-axis denotes the logarithm of the pricing error, which is defined as the absolute value of the difference between the theoretical price and the price calculated by the algorithm. The squares and triangles denote the pricing results for the original Carr and Madan's method and our method, respectively. The parameter values used for the VG model are $S_0 = 100$, $K = 100$, $r = 0$, $T =$ 4 months, $\sigma = 0.1213$, $\nu = 0.1686$, The parameter values used for the Heston model are $S_0 = 100$, $K = 100$, $r = 0$, $T =$ 4 months, $\kappa = 1.49$, $\theta = 0.0671$, $\sigma = 0.742$, $\rho = -0.571$, and $V_0 = 0.0262$. The damping coefficient and the effective upper limit for integration are set to be 1.5 and 1024, respectively, as in Carr and Madan (1999). and $\theta = -0.1436$. The pricing results implied by the original Carr and Madan's method with 2^20 quadrature points are used as the benchmarks.

Table 1: Deep Out-of-the-Money Call Prices.

This table demonstrates the deep out-of-the-money call option prices implied by the Heston model based on the setting $S_0 = 100$, $r = 0.03$, $T =$ 6 months, $\kappa = 2$, $\theta = 0.04$, $\sigma = 0.5$, $\rho = -0.7$, and $V_0 = 0.04$, with strike prices ranging from 160 to 200. The damping coefficient and the effective upper limit for integration are set to be 1.5 and 1024, respectively, as in Carr and Madan (1999). The second column lists the number of grids used for quadrature.

		Carr and Madan		Ours	
Strike	Number of Grids	Time	Call Price	Time	Call Price
160	2^{11}	1.139	-0.00261254	1.357	0.00006875
170	2^{11}	1.123	-0.00266961	1.357	0.00001118
180	2^{11}	1.170	-0.00267840	1.358	$1.89\text{E}-06$
190	2^{11}	1.139	-0.00267954	1.341	$2.51\text{E}-07$
200	2^{12}	2.293	$-1.35\text{E}-07$	2.637	$8.23\text{E}-08$

We demonstrate this result by the Heston model with the same parameters used in Carr and Madan (2009). Table 1 shows that under the same number of quadrature grids, our method returns reasonable prices with about only 20 percent time increment, but the original Carr and Madan's method still returns negative values. Since the FFT is more efficient for number of grids that is a power of 2, we have to double the size of the FFT when a higher accuracy level is required. Therefore the 20 percent time increment is acceptable since the total computational time is approximately doubled if we double the number of quadrature grids.

5 Conclusions

This paper provides an error-reducing method for the option valuation method studied by Carr and Madan (1999). Under the same computational time constraint, our method improves the pricing accuracy significantly. It is because of this that the integrand in the Fourier inversion is smoother and has smaller amplitude for our method. In this work, we illustrate the benefits of our methodology by vanilla option only; however, our method can be extend to other family of payoffs with a suitable proxy term. On the other hand, the proxy term of the call price doesn't necessarily have to be the fair price of some kinds of financial assets; a function with analytic Fourier transform could be the candidate. We also anticipate that the idea in this paper can be further extend to other kinds of numerical methods.

References

Bakshi, G. and D. Madan (2000). Spanning and derivative-security valuation. *Journal of Financial Economics 55*(2), 205–238.

Black, F. and M. Scholes (1973). The pricing of options and corporate liabilities. *The Journal of Political Economy*, 637–654.

Carr, P., H. Geman, D. Madan, and M. Yor (2002). The fine structure of asset returns: An empirical investigation. *Journal of Business 75*(2), 305–332.

Carr, P. and D. Madan (1999). Option valuation using the fast fourier transform. *Journal of Computational Finance 2*(4), 61–73.

Carr, P. and D. Madan (2009). Saddlepoint methods for option pricing. *The Journal of Computational Finance 13*(1), 49–61.

Heston, S. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies 6*(2), 327–343.

Madan, D., P. Carr, and E. Chang (1998). The variance gamma process and option pricing. *European Finance Review 2*(1), 79–105.

Merton, R. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics 3*(1-2), 125–144.